

— PIPELINE DEMO — 2026-05-18

From clinical evidence to computable FHIR, end to end.

A walkthrough of the ReasonHub informatics pipeline applied to **CMS165 — Controlling High Blood Pressure (NQF 0018)**. Starting from raw sources and ending with a deployable FHIR R4 package, with every step reviewer-gated and fully traceable.

Three lifecycle stages. Two tools. Three human review checkpoints.



STAGE 1 — L1 DISCOVERY & INGEST

BK Brian Kaney Researcher

Build the full CMS165 hypertension quality measure pipeline — evidence base, structured artifacts, and computable FHIR output with CQL. Start from scratch.

RH pi agent rh-inf-discovery skill

Scaffolding the topic and running discovery. I'll produce a curated source plan across six categories relevant to CMS165.

• `rh-skills init hypertension-cms-measure`

`rh-skills CLI`

```
rh-skills init hypertension-cms-measure \
  --domain hypertension --author "Brian Kaney"
```

```
topic created → topics/hypertension-cms-measure/
event logged: topic_created (2026-05-18T19:43:08Z)
```

Discovery plan produced — 18 sources identified. Downloading all open-access sources now.

• rh-skills ingest normalize + annotate

rh-skills CLI

rh-skills ingest normalize hypertension-cms-measure
rh-skills ingest annotate hypertension-cms-measure

13 of 18 sources downloaded (5 unavailable: HTTP 403/404)
normalized → sources/normalized/*.md
annotated → 75 concept annotations across 13 sources
uspstf: 9 | cdc: 7 | gbd: 8 | vsac: 7 | qi-core: 8
gravity: 8 | pmid-41439541: 9 | pmid-41186866: 8 | nct: 5
icd-10-cm: 4 | snomed: 4 | loinc: 4 | rxnorm: 6

5 sources unavailable: CMS165 eCQM page (HTTP 404), ACC/AHA 2017 guideline (HTTP 403 — AHA blocks automated access), HCUP statistical brief (HTTP 404). Failures logged in `discovery-plan.yaml` for manual follow-up. Pipeline continued with the 13 successful downloads.

13 sources ingested, normalized to Markdown, classified, and annotated. Each source now carries YAML front matter with type, evidence level, domain tags, and concept annotations. Checksums recorded in `tracking.yaml`.

SOURCE	TYPE	EVIDENCE LEVEL
USPSTF Hypertension Screening 2021	Guideline	USPSTF-B
CDC NCHS Hypertension Surveillance	Government data	N/A
GBD 2023 — Global Burden of Disease	Health economics	Ia
VSAC NQF 0018 Value Sets	Value set	Reference standard
QI-Core FHIR IG	FHIR IG	Reference standard
Gravity SDOH FHIR IG	FHIR IG	Reference standard
PMID 41439541 — Medicare HTN QI	PubMed article	Grade B
PMID 41186866 — Patient Activation RCT	PubMed article	Ib
NCT07556783 — Telephone HBPM Trial	ClinicalTrials.gov	N/A
ICD-10-CM, SNOMED CT, LOINC, RxNorm	Terminology ×4	Reference standard

STAGE 2 – L2 EXTRACT (STRUCTURED ARTIFACTS)

 **pi agent** rh-inf-extract skill

Planning the extraction. Before generating structured artifacts, I'll run the **concept review** — each of the 75 annotated concepts is looked up against live terminology systems via **ReasonHub MCP**, then presented for your approval.

• reasonhub__search_snomed

ReasonHub MCP

query: "essential hypertension disorder"

59621000	Essential hypertension	score: 0.97
38341003	Hypertensive disorder, systemic arterial	score: 0.95
429457004	Systolic essential hypertension	score: 0.91
170578008	Poor hypertension control	score: 0.88
56218007	Systolic hypertension	score: 0.86

• reasonhub__search_loinc

ReasonHub MCP

query: "blood pressure systolic diastolic panel vital sign"

85354-9	Blood pressure panel with all children optional	score: 0.98
8480-6	Systolic blood pressure	score: 0.97
8462-4	Diastolic blood pressure	score: 0.97
55284-4	Blood pressure systolic and diastolic	score: 0.91 [DISCOURAGED]

• reasonhub__search_icd10

ReasonHub MCP · SDOH codes

query: "food insecurity housing instability transportation financial social determinants"

Z59.41	Food insecurity	score: 0.96
Z59.81	Housing instability, housed	score: 0.94
Z59.82	Transportation insecurity	score: 0.95
Z59.86	Financial insecurity	score: 0.92
Z60.4	Social exclusion and rejection	score: 0.88

• reasonhub__search_rxnorm

ReasonHub MCP

query: "antihypertensive drug class ACE inhibitor ARB calcium channel blocker thiazide diuretic"

968	Antihypertensive agent	score: 0.97
1430745	ACE INHIBITORS, COMBINATIONS	score: 0.94
3571	Thiazide diuretic	score: 0.96
1899	Product containing calcium channel blocker	score: 0.95
691779	BETA BLOCKERS/RELATED	score: 0.93
691944	ANGIOTENSIN II INHIBITOR	score: 0.94

Concept review complete. 75 concepts reviewed across 13 sources — **43 approved with standardized codes** across SNOMED CT, ICD-10-CM, LOINC, and RxNorm. Written to `structured/concepts.yaml`.

Extract plan generated with 6 proposed artifacts. Presenting for review.

rh-skills promote extract plan

rh-skills CLI

rh-skills promote extract plan hypertension-cms-measure

wrote: topics/hypertension-cms-measure/process/plans/extract-plan.yaml
wrote: topics/hypertension-cms-measure/process/plans/extract-plan-readout.md
6 artifacts proposed from 13 sources

ARTIFACT	SOURCES	PURPOSE	VERDICT
measure	All 13	CMS165 populations — initial pop, denominator, exclusions (ESRD N18.6, hospice, frailty), numerator (BP < 140/90)	Approved
terminology	VSAC, LOINC, SNOMED, ICD-10-CM, RxNorm	7 value sets + concept map for all CMS165 terminology	Approved
decision-table	USPSTF, QI-Core, VSAC, Gravity, CDC, NCT trial	10 ECA rules across HTN screening, BP monitoring, SDOH referral, QI outreach	Approved
evidence-summary	CDC, GBD 2023, Gravity, PMID 41439541, NCT trial	8 graded findings on prevalence, global burden, QI interventions, SDOH drivers	Approved
assessment	Gravity SDOH IG, USPSTF	9-item instrument: 5 SDOH boolean screens + 4 BP measurements with LOINC + UCUM	Approved
policy	NCT trial only	Trial registry entry — not a coverage/authorization policy source	Rejected

BK Brian Kaney Reviewer — extract plan approval

Approved. The policy rejection is correct — NCT trial evidence belongs in the evidence-summary. Proceed with implementation.

RH pi agent implementing L2 artifacts

rh-skills promote approve + implement

rh-skills CLI

rh-skills promote approve hypertension-cms-measure
rh-skills promote implement hypertension-cms-measure

status: approved (reviewer: Brian Kaney, 2026-05-18T20:08:17Z)
deriving structured artifacts...
measure → structured/measure/measure.yaml 20:14:00Z
terminology → structured/terminology/terminology.yaml 20:17:04Z
decision-table → structured/decision-table/decision-table.yaml 20:12:39Z
evidence-summary → structured/evidence-summary/evidence-summary.yaml 20:15:11Z
assessment → structured/assessment/assessment.yaml 20:16:04Z
policy → REJECTED — skipped
5 artifacts written, 1 skipped

Five L2 structured artifacts written. Each records its derived-from sources and carries full evidence traceability claims. A snapshot of the measure specification:

```
# structured/measure/measure.yaml (excerpt)
title: "CMS165 – Controlling High Blood Pressure: Quality Measure Specification"

populations:
  - type: initial-population
    description: Patients aged 18–85, qualifying outpatient encounter,
                 active essential hypertension (I10 / SNOMED 59621000)
                 diagnosed ≥91 days before end of measurement period

  - type: denominator-exclusion
    description: ESRD (N18.6), hospice, frailty + advanced illness,
                 pregnancy (O-codes), death during measurement period

  - type: numerator
    description: Most recent BP reading:
                 systolic < 140 mm[Hg] (LOINC 8480–6)
                 diastolic < 90 mm[Hg] (LOINC 8462–4)

scoring:
  method: proportion
  reporting_period: calendar year (Jan 1 – Dec 31)

stratification:
  - by: race_ethnicity    # required 2024+
  - by: age_group         # [18–39, 40–64, 65–85]
  - by: payer_type        # [Medicare, Medicaid, Commercial]
```

STAGE 3 – L3 FORMALIZE (FHIR R4 + CQL)

 **pi agent** rh-inf-formalize skill

Mapping each L2 artifact to FHIR R4 resource types via named formalization strategies. Before writing any JSON, I'll verify all codes via ReasonHub MCP.

- reasonhub__codesystem_verify_code – LOINC 85354–9

ReasonHub MCP

code: "85354–9" system: "http://loinc.org"

valid: true
display: "Blood pressure panel with all children optional"
status: ACTIVE
EXAMPLE_UCUM_UNITS: mm[Hg]

• reasonhub__codesystem_subsumes – SNOMED hierarchy check

ReasonHub MCP

```
code_a: "38341003" // Hypertensive disorder
code_b: "59621000" // Essential hypertension
system: "http://snomed.info/sct"
```

result: subsumes

38341003 (Hypertensive disorder) IS-A parent of 59621000 (Essential hypertension)
Both are valid for the SNOMED denominator value set.

• reasonhub__valueset_expand – SDOH Z59.x composition

ReasonHub MCP

```
system: "http://hl7.org/fhir/sid/icd-10-cm"
concepts: Z59.41, Z59.81, Z59.811, Z59.82, Z59.86, Z60.4
```

expansion.total: 6 – all codes valid and active

All codes verified. Formalize plan generated for review:

• rh-skills promote formalize plan

rh-skills CLI

```
rh-skills promote formalize plan hypertension-cms-measure
```

```
wrote: topics/hypertension-cms-measure/process/plans/formalize-plan.yaml
5 strategies proposed | primary implementation target: measure
```

STRATEGY	L2 INPUT	FHIR RESOURCES
terminology	terminology.yaml	ValueSet x8 + ConceptMap x1
assessment	assessment.yaml	Questionnaire (9 items — LOINC + SNOMED + ICD-10 + UCUM units)
evidence-summary	evidence-summary.yaml	Evidence + EvidenceVariable + Citation
decision-table	decision-table.yaml	PlanDefinition (eca-rule, 10 rules) + Library (CQL)
measure — primary target	measure.yaml	Measure (proportional, 4 populations, 3 strats) + Library (CQL + ELM)

BK Brian Kaney Reviewer — formalize plan approval

Approved. **measure** is the correct primary target for CMS165. Generate everything.

RH pi agent rh-inf-cql skill — validate, compile, eval

CQL compilation uses two CLI layers. `rh-skills cql` is a project-aware wrapper that resolves topic paths; `rh` is the underlying CLI that performs the actual compilation and evaluation.

Step 1 — Validate

- rh cql validate

rh CLI

```
# Project-aware wrapper
rh-skills cql validate hypertension-cms-measure MeasureLibrary
rh-skills cql validate hypertension-cms-measure DecisionTableLibrary
```

```
# Underlying rh command
rh cql validate topics/hypertension-cms-measure/computable/MeasureLibrary.cql
```

```
MeasureLibrary.cql – 0 errors, 0 warnings
DecisionTableLibrary.cql – 0 errors, 0 warnings
```

Step 2 — Compile to ELM JSON

- rh cql compile

rh CLI

```
# Project-aware wrapper
rh-skills cql translate hypertension-cms-measure MeasureLibrary
```

```
# Underlying rh command
rh cql compile topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  --output topics/hypertension-cms-measure/computable/elm/MeasureLibrary.json
```

```
# With debug annotations (adds source locators + result types to ELM nodes)
rh cql compile topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  --output topics/hypertension-cms-measure/computable/elm/MeasureLibrary.json \
  --debug
```

```
# Strict mode – disables implicit conversions
rh cql compile topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  --output topics/hypertension-cms-measure/computable/elm/MeasureLibrary.json \
  --strict
```

```
wrote: computable/elm/MeasureLibrary.json
wrote: computable/elm/DecisionTableLibrary.json
```

Step 3 — Evaluate expressions against fixture data

• rh cql eval

rh CLI

```
# Evaluate a single named define against a FHIR Bundle
rh cql eval \
  topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  "Numerator" \
  --data tests/cql/MeasureLibrary/case-001-controlled-bp/input/bundle.json

# With step-by-step trace (useful for debugging null propagation)
rh cql eval \
  topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  "Most Recent Systolic Value" \
  --data tests/cql/MeasureLibrary/case-002-uncontrolled/input/bundle.json \
  --trace

# With additional library include path (e.g. for FHIRHelpers)
rh cql eval \
  topics/hypertension-cms-measure/computable/MeasureLibrary.cql \
  "Initial Population" \
  --lib-path topics/hypertension-cms-measure/computable/ \
  --data bundle.json
```

```
{"Numerator": true}
{"Most Recent Systolic Value": 128.0}  // trace: 14 evaluation steps
```

Additional `rh cql` commands for authoring and debugging:

• rh cql explain / repl / download

rh CLI

```
# Parse tree inspection
rh cql explain parse  topics/hypertension-cms-measure/computable/MeasureLibrary.cql

# Semantic analysis details
rh cql explain compile topics/hypertension-cms-measure/computable/MeasureLibrary.cql

# Interactive REPL
rh cql repl

# Fetch FHIRHelpers for local validation (runtime dependency – not committed to computable/)
rh download package fhir.cqf.common 4.0.1
```

FHIRHelpers-agnostic engine. The `rh` evaluator does not auto-inject `FHIRHelpers.ToConcept()` calls — unlike the Java reference translator. Every library that uses FHIR coded, quantity, or date types must include it explicitly:

```
include fhir.cqf.common.FHIRHelpers version '4.0.1' called FHIRHelpers
```

COMMAND	WHAT IT DOES	KEY FLAGS
<code>rh cql validate <file.cql></code>	Syntax and semantic check	—

COMMAND	WHAT IT DOES	KEY FLAGS
<code>rh cql compile <file.cql></code>	Compile to ELM JSON	<code>-o / --output</code> , <code>--debug</code> , <code>--strict</code>
<code>rh cql eval <file.cql> <name></code>	Evaluate one named expression	<code>--data <bundle.json></code> , <code>--lib-path <dir></code> , <code>- -trace</code>
<code>rh cql explain parse</code>	Print parse tree	—
<code>rh cql explain compile</code>	Print semantic analysis	—
<code>rh cql repl</code>	Interactive expression REPL	—
<code>rh download package <id> <ver></code>	Fetch a FHIR package locally	—

RH **pi agent** rh-inf-formalize + rh-inf-cql skills

- rh-skills promote formalize-approve + implement

rh-skills CLI

```
rh-skills promote formalize-approve hypertension-cms-measure
rh-skills promote implement hypertension-cms-measure --stage formalize

status: approved (reviewer: bkaney, 2026-05-18T21:19:34Z)
formalizing...
  terminology      → 9 resources (ValueSet ×8 + ConceptMap) 21:22:07Z
  assessment       → 1 resource  (Questionnaire)           21:22:11Z
  evidence-summary → 3 resources (Evidence+EvidenceVariable+Citation) 21:22:14Z
  decision-table   → 2 resources (PlanDefinition+Library)    21:22:22Z
  measure          → 2 resources (Measure+Library)          21:22:25Z
CQL compiled → ELM JSON for MeasureLibrary and DecisionTableLibrary
```

The generated CQL library implements the CMS165 population logic. It handles both panel observations (`LOINC 85354-9`) and standalone component codes, extracts typed quantities from `FHIR.Quantity`, and evaluates the `< 140/90 mmHg` threshold:

```
/**
 * MeasureLibrary v1.0.0 – CMS165 Controlling High Blood Pressure
 * NQF 0018 | MIPS #236 | HEDIS CBP
 */
library MeasureLibrary version '1.0.0'
using FHIR version '4.0.1'

valueset "Essential Hypertension ICD10":
  '...ValueSet/vs-htn-dx-icd10'

valueset "Blood Pressure Observations LOINC":
  '...ValueSet/vs-bp-obs-loinc'

parameter "Measurement Period" Interval<DateTime>
  default Interval[@2024-01-01, @2024-12-31]

define "Initial Population":
  "Age 18 to 85"
  and exists "Qualifying Outpatient Encounters"
  and "Has Active Hypertension Diagnosed Before October"

define "Denominator Exclusion":
  "Has End Stage Renal Disease" // ICD-10-CM N18.6

define "Numerator":
  "Denominator"
  and "Most Recent Systolic Value" < 140
  and "Most Recent Diastolic Value" < 90
```

FINAL – PACKAGE

• rh-skills package hypertension-cms-measure

rh-skills CLI

rh-skills package hypertension-cms-measure

@reason/hypertension-cms-measure v1.0.0
FHIR 4.0.1 | deps: us-core@6.1.0 crmi@1.0.0 cql@2.0.0

Measure-measure.json
Library-measure-library.json (+ MeasureLibrary.cql + elm/MeasureLibrary.json)
PlanDefinition-decision-table.json
Library-decision-table-library.json (+ DecisionTableLibrary.cql)
ValueSet x8
ConceptMap-terminology-conceptmap.json
Questionnaire-assessment.json
Evidence-evidence-summary.json
EvidenceVariable-evidence-summary-evidencevariable.json
Citation-evidence-summary-citation.json
ImplementationGuide-hypertension-cms-measure.json
package.json

22 resources | package: topics/hypertension-cms-measure/package/
event logged: package_created (2026-05-18T21:32:30Z)

Pipeline complete. Final numbers:

13

Sources
ingested

43

Concepts
coded &
approved

5

L2 artifacts (of
6 proposed)

22

FHIR R4
resources

2

CQL libraries +
ELM

3

Human review
checkpoints

Traceability example:

USPSTF 2021 HTML → normalized to Markdown → 9 concepts annotated → measure.yaml (claim et-m-01:
BP < 140/90 threshold) → Measure-measure.json → MeasureLibrary.cql ("Most Recent Systolic
Value" < 140) → @reason/hypertension-cms-measure v1.0.0

HOW THE TOOLS WORK TOGETHER

Two distinct systems drive the pipeline. The rh-skills CLI orchestrates the lifecycle — scaffolding, planning, reviewing, and packaging. ReasonHub MCP provides live terminology intelligence at the two points where coded clinical concepts must be resolved: concept enrichment during extraction, and code verification during formalization.

• **rh-skills CLI**

- › Scaffold topic, download and normalize sources
- › Generate discovery, extract, and formalize plans
- › Present plans to human reviewer; record approve/reject decisions
- › Derive L2 structured YAML artifacts from normalized sources
- › Generate CQL source from L2 structured artifacts
- › Package FHIR IG bundle with dependencies
- › Maintain checksums and event log `tracking.yaml` in

• **ReasonHub MCP**

- › Semantic search across SNOMED CT, LOINC, RxNorm, ICD-10-CM, UCUM
- › Verify individual codes are valid and active in live code systems
- › Retrieve UCUM units for quantitative LOINC observations
- › Check SNOMED subsumption relationships
- › Expand and validate FHIR ValueSet compositions
- › Return similarity-ranked results with display names
- › Surface discouraged codes (e.g., LOINC 55284-4)

• **rh CLI**

- › Validate CQL syntax and semantics — `rh cql validate <file.cql>`
- › Compile CQL to ELM JSON — `rh cql compile --output --debug --strict`
- › Evaluate named expressions against FHIR Bundles — `rh cql eval --data --trace`
- › Parse tree and semantic analysis — `rh cql explain parse/compile`
- › Interactive CQL REPL — `rh cql repl`
- › Fetch FHIR packages for local validation — `rh download package`